TAHIR AHMAD BIJRAN

+91 95966 98567 | tahirbijran@gmail.com | tahirbijran@cukashmir.edu.in
Ganderbal, Jammu & Kashmir, India • Portfolio: bijrantahir.github.io • ORCID: 0009-0002-7115-4499

PROFESSIONAL PROFILE

B.Sc. (Hons.) with Research in Biotechnology student at the Central University of Kashmir (2024–2028, expected), with interdisciplinary interests in molecular and cellular biology, nanobiotechnology, functional nanomaterials, bioinformatics and computational biology. Research Intern at the Functional Nanomaterials & Devices Lab, Central University of Kashmir. Building practical laboratory experience alongside computational and digital skills, with an interest in applying reproducible, data-informed approaches to biological research.

EDUCATION

B.Sc. (Hons.) with Research in Biotechnology | Central University of Kashmir
2024–2028 | Department of Biotechnology

RESEARCH EXPERIENCE

Research Intern | Functional Nanomaterials & Devices Lab (nanomatlab.in)
Central University of Kashmir, Ganderbal, Jammu & Kashmir | August 2026–Present

- Research exposure in a laboratory focused on functional nanomaterials and device-oriented research.
- Developing research skills in scientific literature review, research documentation, experimental planning and interpretation of scientific information.
- Interested in the relationship between material properties, characterization outcomes and potential applications in biotechnology and sensing.

Laboratory website: https://www.nanomatlab.in/

SELECTED PROJECTS & INDEPENDENT WORK

Lab Suite Pro | Independent scientific web application | https://bijrantahir.github.io/LabSuitePro/

- Developed an independent web-based scientific workspace to organise and present laboratory and biotechnology-related resources.
- Demonstrates initiative in digital development, scientific resource organisation and connecting biotechnology workflows with accessible web tools.

COMPUTATIONAL & DIGITAL SKILLS

- Bioinformatics platforms and resources: Galaxy, NCBI resources and BLAST.
- Cloud and computational environments: Google Colab, Kaggle and Termux.
- Programming and web technologies: Python, HTML, CSS and JavaScript.
- Research and development workflow: GitHub, GitHub repositories, version-controlled project presentation and GitHub Pages.
- Computational interests: sequence analysis, biological data handling, genomic data analysis, scientific visualisation and reproducible workflows.

LABORATORY & TECHNICAL SKILLS

Experience in biotechnology laboratory practicals. Key techniques and areas include:

- Blood grouping: ABO and Rh typing using Anti-A, Anti-B and Anti-D sera; observation and interpretation of agglutination.
- Molecular biology: DNA/RNA isolation and handling, agarose gel electrophoresis, preparation and handling of biological samples.
- Microscopy and cell biology: light microscopy, slide preparation, specimen observation and basic cell/ tissue examination.
- Microbiology: aseptic practices, media preparation, microbial inoculation and culture handling, staining and microscopic examination.
- Biochemistry: preparation of reagents and buffers, qualitative biochemical tests, spectrophotometric measurements and basic biomolecule assays.
- General laboratory practice: micropipetting, serial dilution, centrifugation, solution preparation, laboratory safety, equipment handling and scientific record keeping.

SCIENTIFIC WEBINARS, WORKSHOPS & PROFESSIONAL DEVELOPMENT

Attended webinars and workshops across multiple areas of life sciences and emerging technologies, including:

- Next-Generation Sequencing (NGS) and Genomic Data Analysis
- Bioinformatics and Computational Biology
- Molecular Biology and Genetic Engineering
- CRISPR-Cas Genome Editing and Gene Editing Technologies

- Nanotechnology and Nanobiotechnology
- Cell Culture Techniques and Advanced Cell Biology
- Artificial Intelligence and Machine Learning in Biotechnology

RESEARCH INTERESTS

Functional nanomaterials • Nanobiotechnology • Carbon quantum dots • Molecular biology • Cell biology • Genetics and genomics • Cancer Biology • Biochemistry • Enzymology • NGS and genomic data analysis • Bioinformatics • Computational biology • Biomedical biotechnology • AI/ML applications in life sciences

PROFESSIONAL LINKS

- Personal portfolio: https://bijrantahir.github.io/
- Lab Suite Pro: https://bijrantahir.github.io/LabSuitePro/
- NanoMatLab: https://www.nanomatlab.in/
- GitHub: https://github.com/BijranTahir
- LinkedIn: https://www.linkedin.com/in/tahir-ahmad-bijran-183389273/
- ORCID: https://orcid.org/0009-0002-7115-4499

REFERENCES

Dr Anil Chandra

Assistant Professor | Department of Biotechnology

Central University of Kashmir | anil@cukashmir.ac.in

Dr. Dharmendra Singh

HoD & Associate Professor | Department of Biotechnology

Central University of Kashmir | dsingh@cukashmir.ac.in